OFFICIAL CURRICULUM PROPOSAL

Data Science & AI: From Zero to Pro

Comprehensive Masterclass with Production-Grade End-to-End Projects

TARGET TRACK	APPROACH	OUTCOME
Beginner (Level 0) to Pro	Theory + Real-World Deployment	Industry-Ready Portfolio

PROGRAM OVERVIEW

Bridging Theory with Production-Level Deployment

Complete Progression

Takes students from basic Python arithmetic all the way to advanced Deep Learning, LLMs, and MLOps deployment.

End-to-End Projects

Every module culminates in a real-world enterprise project covering data ingestion, modeling, and API containerization.

Career Acceleration

Equips students with GitHub-ready production code and interview-grade machine learning system design experience.

LEARNING OUTCOMES

Core Competencies Developed Across the Track

Advanced Data Manipulation

Clean, merge, transform, and analyze massive unstructured and structured datasets using Pandas, NumPy, and SQL.

Machine Learning Modeling

Build, tune, and evaluate Scikit-Learn models, gradient boosters (XGBoost/LightGBM), and Deep Neural Networks.

MLOps & Model Deployment

Containerize models with Docker, build REST APIs using FastAPI, and deploy scalable solutions to cloud infrastructure.

Modern AI & Generative Models

Implement NLP pipelines, Transformer architectures, Computer Vision, and RAG systems using vector databases.

AUDIENCE & REQUIREMENTS

Who Should Enroll in This Program?

Ideal Candidates

Diverse Professional Backgrounds

- STEM Students seeking high-demand career paths
- Software Engineers transitioning into AI & Data
- Analysts wanting to upgrade from BI to Predictive Modeling
- Beginners with strong logical aptitude starting at Level 0

Prerequisites

Entry Requirements

- Basic computer literacy and logical thinking
- High school level algebra and basic statistics
- Zero coding experience required (Level 0 covers Python)
- High motivation for hands-on project building

CURRICULUM ARCHITECTURE

5 – Stage Masterclass Roadmap

STAGE 01

Level 0: Python & Math

Syntax, Data Structures, OOP, Linear Algebra & Stats.

STAGE 02

Level 1: EDA & SQL

NumPy, Pandas, Data Cleaning, Advanced SQL, Seaborn.

STAGE 03

Level 2: Classical ML

Scikit-Learn, Regression, Classification, XGBoost, Tuning.

STAGE 04

Level 3: Deep Learning

PyTorch, CNNs, RNNs, Transformers, NLP & LLMs.

STAGE 05

Pro: MLOps & Cloud

FastAPI, Docker, CI/CD, MLflow, AWS/GCP Deployment.

FOUNDATION MODULES

Levels 0 & 1: Foundations, EDA & SQL

</> Level 0: Python & Math for AI

- Python syntax, control flow, functions, and OOP
- Essential Math: Vectors, Matrices, Derivatives, Probability
- Environment setup: Jupyter Notebooks, Git & GitHub
- **Project 1:** Automated CLI Data Parsing & Sanitizer

Level 1: Data Wrangling & SQL

- NumPy vectorization & Pandas dataframe manipulation
- Data cleaning, missing value imputation, outlier handling
- Advanced SQL: Joins, Window Functions, CTEs in PostgreSQL
- **Project 2:** Exploratory Data Analysis & Dashboarding on E-Commerce Dataset

CORE MACHINE LEARNING

Level 2: Classical Machine Learning & Feature Engineering

Key Topics Covered

- Supervised Learning: Linear/Logistic Regression, Decision Trees
- Ensemble Methods: Random Forests, XGBoost, LightGBM
- Unsupervised Learning: K-Means Clustering, PCA, Dimensionality Reduction
- Model Evaluation, Cross-Validation, Hyperparameter Tuning (Optuna)

End-to-End Project 3

Predictive House Pricing & Customer Churn Engine: Building a complete machine learning pipeline with automated feature engineering, model selection, evaluation metrics (ROC-AUC, RMSE), and serialization using Scikit-Learn.

ADVANCED AI & DEEP LEARNING

Level 3: Deep Learning, Computer Vision & LLMs

Advanced Architectures

- Neural Networks fundamentals using PyTorch (Autograd, Optimizers)
- Computer Vision: Convolutional Neural Networks (CNNs), ResNet transfer learning
- NLP & Transformers: Tokenization, Attention Mechanisms, Hugging Face
- Generative AI: RAG (Retrieval-Augmented Generation) & Vector Databases (Pinecone/Chroma)

End-to-End Project 4

Enterprise Document Q&A Chatbot (RAG System):
Building a production-ready Generative AI application integrating custom PDF knowledge bases, Hugging Face embeddings, vector storage, and an LLM interface.

PRO TRACK & MLOPS

Pro Level: Production Engineering & Cloud Deployment

Production Engineering

- Building high-performance REST APIs with FastAPI & Pydantic
- Containerization with Docker & Docker Compose
- Experiment Tracking & Model Registry with MLflow
- CI/CD Pipelines with GitHub Actions for automated testing

End-to-End Capstone Project

Full-Stack Cloud ML Application:
Deploying a trained machine learning model packaged inside a Docker container onto AWS/GCP, connected to a FastAPI backend and a Streamlit frontend user interface.

PEDAGOGY & DELIVERY

Active Learning & Industry-Standard Mentorship

1. Concept + Code First

Brief theoretical mathematical intuition followed immediately by live coding and hands-on notebook implementation.

2. Code Reviews

Weekly GitHub pull request reviews by senior data scientists to enforce clean coding standards and best practices.

3. Hackathons & Sprints

Real Kaggle-style data challenges where students collaborate to solve messy, real-world predictive modeling problems.

EVALUATION FRAMEWORK

Multi-Tier Student Assessment Strategy

40%

End-to-End Projects

5 major milestone projects evaluated on code quality, model accuracy, and documentation.

30%

Coding Labs & Quizzes

Weekly Python, SQL, and algorithm challenges testing practical data processing speed.

30%

Capstone Defense & Interview

Live technical defense of the final cloud deployment project before a panel of experts.

READY FOR LAUNCH

Empower Your Students With Production AI Skills

Complete syllabus, notebooks, deployment labs, and evaluation frameworks ready for immediate integration into your institution.

Accept Data Science Proposal